higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

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NATIONAL CERTIFICATE

LOGIC SYSTEMS N6

(8080306)

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09:00–12:00

Calculators may be used.

This question paper consists of 7 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
LOGIC SYSTEMS N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Subsections of questions must be kept together.
5. Write neatly and legibly.

NOTICE TO CANDIDATES

The lives and safety of people depend on logic systems which form an integral part of intrinsic safety in equipment. Logic signalling systems prevent trains from colliding. Logic systems keep people safe in aircraft and in hospital intensive-care units.

Standards must be maintained for the safety of the public and safe operation of machinery and workers. Your work, programming and designs, must be properly planned and presented in this diploma-level examination.

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QUESTION 1

1.1 Glitches are unwanted and may cause serious harm in electronic circuits.

Explain a *glitch*. (2)

1.2 Give THREE serious problems which can be caused by a glitch. (3)

1.3 Demonstrate how a glitch can appear in a digital-to-analogue converter by drawing the staircase output waveform of a 4-bit digital-to analogue converter. The staircase output detailed on the vertical axis must indicate increasing analogue values from 0 mV to 5 mV. Clearly show the corresponding 4-bit digital values on the horizontal axis. Assume that only ONE glitch occurs between steps three and four because bit switch-on takes longer than bit switch-off. Draw this glitch superimposed on the staircase waveform. Clearly show the shape and direction of the glitch as it appears on an oscilloscope screen. Enter the momentary value which causes the glitch and show between which values this momentary value appears. (8)

1.4 Below is a program segment with an accompanying instruction set and information about preloaded addresses.

Analyse this given information to answer the questions.

INSTRUCTION LOCATION	INSTRUCTION WORD		DESCRIPTION
	OP CODE	ADDRESS	
AAAA	LOAD	F1B1	Load r
AAAB	ADDA	F1B1	Add r
AAAC	ADDA	F1B1	Add r
AAAD	ADDA	F1B1	Add r
AAAE	ADDA	F1B1	Add v
AAAF	SUBT	F1B2	Subtract s
AAB0	SUBT	F1B2	Subtract s
AAB1	SUBT	F1B2	Subtract s
AAB2	ADDA	F1B3	Add t
AAB3	ADDA	F1B3	Add t
AAB4	ADDA	F1B3	Add t
AAB5	ADDA	F1B3	Add u
AAB6	STOR	F1B8	Store in z (new value)
AAB7	OUTP		Output
AAB8	HALT		Halt

The following instruction set is available:

OP CODE	DESCRIPTION
LOAD	Load accumulator
ADDA	Add to accumulator
SUBT	Subtract from accumulator
MULT	Multiply
BRAN	Branch if the accumulator is negative
BRAP	Branch if the accumulator is positive
STOR	Store
OUTP	Output the contents of the accumulator
HALT	Halt (nonmemory referenced instruction)

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Information about preloaded addresses:

The operands r, s, t, u v, w, x, y and z are stored in consecutive hexadecimal addresses:

Operand r is stored in address F1B1
Operand s is stored in address F1B2
Operand t is stored in address F1B3
Operand u is stored in address F1B4 and so on.

The value of r is preloaded with the decimal integer value 1100
The value of s is preloaded with the decimal integer value 1000
The value of t is preloaded with the decimal integer value 900
The value of u is preloaded with the decimal integer value 800

The integer values preloaded into the alphabetical variables which follow u continue to decrement by 100. Z is reserved to store the end result.

1.4.1 Use an appropriate heading and write down the simplest mathematical form of the equation which this program segment solves. (5)

1.4.2 Write down, under an appropriate heading, what sort of programming language has been used to write this program segment in QUESTION 1.4.1 (Machine Code, BASIC, COBOL, C++, FORTRAN or any other). (2)

1.4.3 Use an appropriate heading and write down the value (numerical or otherwise) printed out on completion of the program segment in QUESTION 1.4.1. Show ALL steps in your calculation. (5)

[25]

QUESTION 2

2.1 Logic systems engineers must be able to use digital-to-analogue converters to control certain types of instrumentation.

Draw a schematic diagram of a weighted resistor-type digital-to-analogue converter that can convert a 3-digit BCD (binary-coded decimal) to an equivalent analogue output voltage. Use an operational amplifier and only one reference voltage. Clearly show the ratios and resistor values of the resistors used in the circuit. The largest resistor in the weighted resistor ladder has a value of 10 mega-ohms and the feedback resistor on the operational amplifier has a value of 10 kilo-ohms.

Show the input switch positions for a digital input of 401 BCD. (15)

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2.2 List the programming languages defined by the International Electro-technical Commission under the Open International Standard IEC 61131-3.

Clearly show the programming format of each language (text or graphical). (10)
[25]

QUESTION 3

3.1 Draw a complete, fully labelled block diagram of a voltage-to-time-to-digital converter. (10)

3.2 Briefly explain the operation of this converter. (5)
[15]

QUESTION 4

4.1 Draw a large, neat, fully labelled diagram of an operational amplifier configured as a comparator. The operational amplifier is used to control the level of liquid in a tank. Show the connections to sensor(s) and actuator(s) necessary to control the level. Show signal directions. Show any analogue-to-digital or digital-to-analogue converters that may be needed. Fully integrate the control circuitry with the objects which are being controlled. (9)

4.2 Draw a large, neat, fully labelled block diagram to show the interfacing of input/output ports of a computer which is used to control the level of liquid in a tank. The labels of the block diagram must appear on the block diagram itself. Clearly show the configuration of each input/output port. All interfacing and process functions needed to monitor and control the level must be shown. Show signal directions. Show the interfacing of sensor(s) and actuator(s). Label the feedback and action loops. (9)

4.3 A programmer is able to program a PLC (programmable logic controller) using either diagram mode or mnemonics (statement list). Clicking on icons gives the programmer the choice of display.

Describe the form of circuit diagram which is displayed to represent a NOR statement. (2)
[20]

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QUESTION 5

5.2 The PLC program segment below is given an instruction list (also commonly called a statement list). The program allows MOTOR_1 to run for ten seconds (#100) before it is disconnected by TIMER_1. The values given in the VALUE column are (with the exception of the hexadecimal timer value) Boolean values (either 0 or 1). The values given in the table are the unenergised values before the circuit is switched on.

Draw the ladder diagram format of this program segment. Place addresses above each component. Place comments under each component.

RUNG	STEP	INSTRUCTION	OPERAND	VALUE	COMMENT
0	0	LD	**A**	0	PUSHBUTTON ON_SWITCH
	1	OR	**Z**	0	MOTOR_1 (CONTACT)
	2	ANDNOT	**T0001**	0	TIMER CONTACT
	3	OUT	**Z**	0	MOTOR_1
1	4	LD	**Z**	0	MOTOR_1 (CONTACT)
	5	TIM	**0001**	0100Hex	TIMER_1
			#100		

(10)

5.2 Explain what the International Electrotechnical Commission (I.E.C.) is and describe its functions and importance in industry. (5)
[15]

TOTAL: 100

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